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Key Points:

- We implement a method to estimate contractional fault strain independent of the number of faults considered
- We resolve the long-standing disagreement of the amount of Mercury's global contraction
- Our results indicate Mercury radially contracted by multiple kilometers for a wide range of fault geometries and depths

Supporting Information:

Supporting Information may be found in the online version of this article.

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Several Kilometers of Global Contraction on Mercury: A Sample-Size Independent Assessment of Fault Strain

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Abstract Mercury underwent global contraction due to the sustained cooling of the planet. Positive-relief landforms, found widespread across Mercury, are thought to be the surface expressions of thrust faults accommodating the contraction. Disagreement exists in the literature on the amount of contraction, with estimates of radius change ranging from ~1 to 7 km. These differences solely arise from the method used to estimate the fault population strain, which relies on the number of structures. Here, we adapt a previous framework by which the continuum approximation to shortening strains can be determined from fault length and displacement statistics for an incompletely sampled fault population. We apply this method to three data sets that sample different numbers of faults. Our results show that even for conservative fault parameters, 2 to 3.5 km of radial contraction are returned, irrespective of the data set used, and thus resolve the debate on the amount of global contraction on Mercury.

Plain Language Summary The planet Mercury has shrunk due to the long, sustained cooling of its interior. This process has led to the development of faults to accommodate the shrinking in the cold uppermost rocky parts of the planet. Multiple studies use the uplifted topography that overlie these faults to estimate the amount Mercury has shrunk. However, these studies are in disagreement with one another due to the number of landforms that each study attributes to global contraction. Here, we use an alternate approach of calculating the change in volume caused by a population of faults by statistically scaling the change from the largest fault to the entire fault population. From the literature, we use three fault data sets with noticeably different numbers of structures that previously showed the disagreement. The method in this study produces results that are comparable to one another, resolving the long-standing disagreement in the literature. Our results find that at least three kilometers of change in Mercury's radius occurred for a conservative set of fault-geometric parameters. Therefore, this newly applied method should replace the previous approach to estimate the amount by which Mercury has shrunk.

1. Introduction: Disagreement on Mercury's Radial Contraction

Thermal evolution models and observations of Mercury's surface indicate that the planet has undergone global contraction, a process caused by long, sustained planetary cooling (e.g., Solomon, 1978). Shortening strain from global contraction is widely accepted to be accommodated in the lithosphere via thrust faulting, which manifests at the surface as positive-relief landforms (Byrne et al., 2014, 2018; Solomon et al., 2008; Strom et al., 1975). Previous studies measured the length and relief of these shortening landforms to estimate the total contractional strain of Mercury (Byrne et al., 2014; Di Achille et al., 2012; Watters & Nimmo, 2010; Watters et al., 1998). All of these previous studies rely on the same method, by which a map of thrust fault related landforms is produced, lengths of the landforms are extracted, relief of a subset of structures is measured and assumed to be related to fault displacement via the fault dip such that a displacement-to-length relationship is established, which then is extrapolated to all mapped structures. All of these studies sample from the same population of thrust faults. Strain is then calculated for the area of the faulted domain by summing the strain of each of the mapped structures. This method is thus dependent on the number of structures considered.

The dependency on the number of faults produced a discrepancy in radius change estimates in the literature. Watters (2021), who used $n = 653$ faults, calculated Mercury's radius to have decreased by 0.9–1.3 km, whereas Byrne et al. (2014), who considered a population of faults containing $n = 5,934$ structures, determined a range of 3.1–7.1 km of radius change. Watters (2021) assumed that shortening landforms in Mercury's smooth plains units are attributed to lithospheric loading and associated subsidence from the emplacement of the volcanic units rather than global contraction and thus excluded the structures and area covered by plains units from the calculation.

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Byrne et al. (2014) considered a wider range of possibilities, including and excluding the smooth plains units. In their work, radial shortening was calculated using shortening landforms across eight great circles and by using the displacement length scaling. In all cases, with or without the shortening landforms in the smooth plains units, Byrne et al. (2014) mapped more faults within the same areas that were later considered by Watters (2021). Therefore, the method used in these studies is a disadvantage, as different geological interpretations result in a different number of considered structures, and in turn, cause the discrepancy in estimates of the amount of global contraction despite sampling the same population of faults.

An additional component of radius change was proposed to occur prior to the onset of faulting as the lithosphere may resist faulting until its strength is reached. Klimczak (2015) estimates the strength of Mercury's dry lithosphere by using rock mechanical considerations for a volume of rock with a range of degrees of fracturing that have been developed and widely used for calculating the frictional resistance of sliding along faults on Earth. This strength was then compared to the horizontal compressive stresses produced by global contraction given in Melosh and McKinnon (1988). Klimczak (2015) found that Mercury could have contracted an additional 0.4–2.1 km radially prior to the onset of faulting. This results in radius changes from as little as 1.3 km, when adding the lowest possible amounts from Watters (2021) and Klimczak (2015), to as much as 9.2 km, when considering the upper estimates from Byrne et al. (2014) and Klimczak (2015). This range of values is a critical constraint for thermal evolution models, whereby the most plausible solutions require a radial shortening of ~5–10 km (Breuer et al., 2007; Hauck et al., 2004; Michel et al., 2013; Solomon, 1977; Tosi et al., 2013). Driven by the constraint of low values of radial contraction by Watters et al. (1998), Tosi et al. (2013) identified a very small fraction of model solutions that required less than 3.5 km of radial shortening by assuming very large reference viscosities ($>10^{21}$ Pa s).

The large range of estimated amounts of radial shortening on Mercury that exist in the literature is only a very loose constraint for thermal evolution models, highlighting the need for improvement in the method on which the estimations are based on. Thus, we provide an alternative assessment of Mercury's global contractional strain, for which we apply fault strain theory introduced and tested by Twiss and Marrett (2010a, 2010b) to three published Mercury thrust fault data sets to reevaluate to amount of radial contraction.

2. Methods

The fault strain theory laid out by Twiss and Marrett (2010a) is based on the fact that an entire population of faults is never really fully resolved when estimating the strain of a faulted volume. Twiss and Marrett (2010a) developed this methodology for regions on Earth that have accommodated strain via large fault populations. Instead, this approach utilizes the displacement of the largest fault in the population, which is easiest to detect, and scales it using both displacement-to-length and fault-length-frequency statistics to estimate the total strain of the faulted volume. This method assumes infinitesimal strain, where the faulted volume must be large relative to the dimensions of the largest fault. If, for instance, this method is used to assess regional-scale deformation on planetary bodies, the scale of the volume should be substantially larger than the extent of the largest fault. This assumption is valid for our purposes, as we are considering the entire brittle volume of Mercury's lithosphere. This method also assumes that all structures in the population have the same fault geometric parameters, such as dip and fault shape, which was also assumed in previous works (Byrne et al., 2014; Di Achille et al., 2012; Watters et al., 1998; Watters & Nimmo, 2010). Fault geometries on planetary bodies other than Earth are generally poorly constrained but there is a growing body of work quantifying subsurface structure of shortening landforms (i.e., for Mercury: Egea-Gonzalez et al., 2012; Loveless et al., 2025a; Pegg et al., 2021; Peterson et al., 2020; Watters et al., 2001, 2002). We therefore calculate Mercury's contractional strain for the range of known fault geometries using the method by Twiss and Marrett (2010a), thereby removing the dependence on fault sample size and instead emphasizing the reliance on objective parameters such as physical attributes and fault geometry.

The method by Twiss and Marrett (2010a) allows us to calculate the strain in Mercury's faulted volume of the lithosphere independent of the number of considered faults. We test this approach for three different data sets in the literature that sample substantially different numbers of thrust faults from the same population of structures (Figure 1). We use the previously mentioned data from Byrne et al. (2014) who sample $n = 5,934$ structures, from Watters (2021), who samples $n = 653$ structures, and from Loveless et al. (2024, 2025a, 2025b), in which $n = 100$ structures were considered.

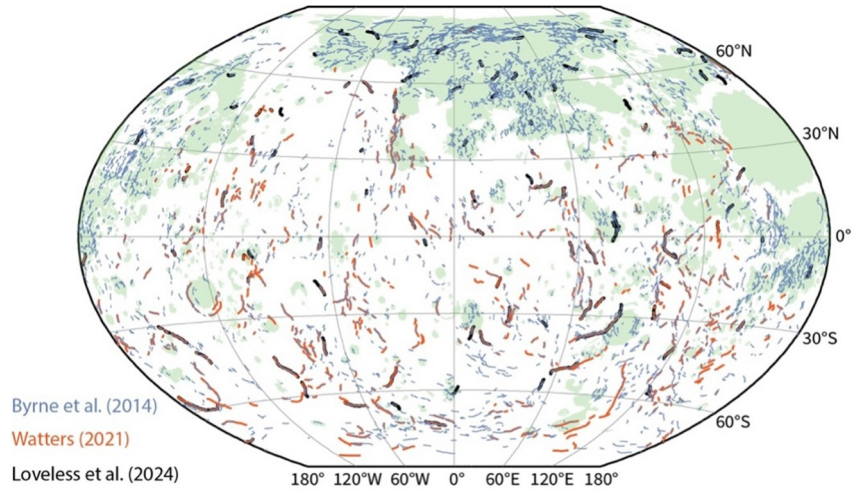


Figure 1. Comparison of mapped thrust fault populations considered in this study across Mercury in Winkel Tripel projection, color-coded based by source. The data set by Byrne et al. (2014) is displayed in blue atop the data set by Watters (2021), which is displayed in orange. The faults from Loveless et al. (2024) are displayed in thick black lines. For reference, smooth plains units (Denevi et al., 2013) are colored in light green.

Twiss and Marrett (2010a) derived the equation for the total strain of faults sampled in three dimensions of a volume in the horizontal direction t :

$$e_{3,t}^{\text{tot}} = e_{3,t}^{(1)} \left(\frac{\left(1 + \frac{2}{p}\right)}{1 - \left(s - \frac{2}{p}\right)} \right), \quad (1)$$

where $e_{3,t}^{(1)}$ is the shortening strain of the fault with the highest displacement in the volume. The parameters p and s are derived from statistics based on the total population of fault parameters (see Section 3: Parameterization for how to obtain values). The strain of fault i in direction t is found by:

$$e_{3,t}^{(i)} = \mathfrak{P}_{3,t}^{(i)} \left[\frac{\delta^{(i)} \cos(\phi)}{T} \right], \quad (2)$$

where $\delta^{(i)}$ is the displacement of the i th fault and ϕ is the angle between the horizontal direction t and the slip direction on the fault (Figure 2), corresponding to subtracting the fault dip, α , from 180° . Here, $\mathfrak{P}^{(i)}$ is the probability for an individual fault to be intersected by a random line parallel to t . The parameter T is the dimensional length of the faulted volume in direction t (Figure 2). The probability for an individual fault written as a continuous function of fault-displacement δ is given by:

$$\mathfrak{P}_{3,t}^{(i)} = \frac{A_f^i \cos(\theta^i)}{A_t} = \frac{\lambda B^2 \cos(\theta)}{A_t} \delta^{(i)2}. \quad (3)$$

Here, A_f is the area of the fault plane. The angle θ is measured between the normal vector to the fault plane and t , or by subtracting α from 90° (Figure 2). The denominator, A_t , is the total cross-sectional area of the faulted volume, which is normal to t . The parameter λ is a geometrical shape factor that accounts for the fault height, as defined by down-dip minor axis length, l , and the length, L , specifying either rectangular or elliptical shapes of the fault tip line. For rectangular fault shapes, λ is simply the ratio of l and L (as shown in Figure 2), whereas for elliptical shapes this ratio is multiplied by $\pi/4$. The factor B is a scaling parameter derived from the displacement-to-length statistic of the faults (see below). The probability in Equation 3 then is the ratio of the area of the fault plane that is projected onto the total cross-sectional area of the faulted volume to the total cross-sectional area

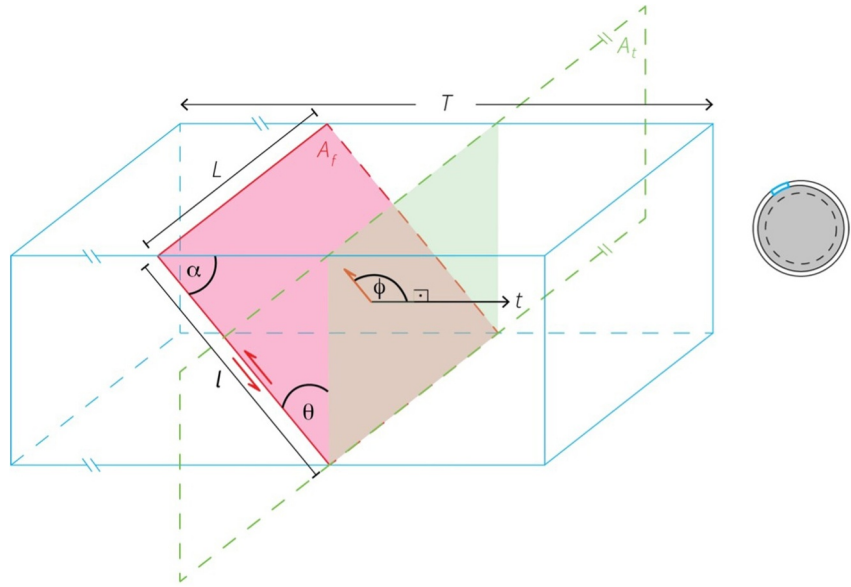


Figure 2. Block diagram of fault geometric parameters modified after Twiss and Marrett (2010a, 2010b). The block (blue outline) is a representative portion of the faulted volume of Mercury's lithosphere. In this diagram the fault plane (A_f) is the area marked as red shaded region and it is here shown as rectangular shape. The plane outlined with green dashes is the total cross-sectional area (A_t) of the faulted volume. The green shaded region represents A_f projected onto A_t . The circle on the right represents Mercury with the white outer portion depicting the planet's lithosphere with exaggerated thickness for clarity. The blue rectangle on the lithosphere corresponds to the portion of the lithosphere shown in the block diagram.

(Figure 2). This probability can be estimated for any fault in a population, as the areas of the fault planes are obtained from the displacement-length and cumulative number-length relationships. Note the absence of the superscript i for the geometric parameters λ and θ on the right-hand side of Equation 3, as these are assumed to be the same for all faults in the population.

The faulted volume, V , is then calculated as the product of A_t and T . We define the faulted volume to be equal to the outer shell of Mercury that has a thickness defined by the depth-extent of the deepest faults, D , and planetary radius, R_M , as:

$$V = A_t T = \frac{4}{3} \pi R_M^3 - \frac{4}{3} \pi (R_M^3 - D^3). \quad (4)$$

Following the approach by Twiss and Marrett (2010a) we combine Equations 1–4, yielding:

$$e_{3,t}^{\text{tot}} = \frac{\lambda B_p^2 (\cos \theta \cos \phi)}{V} (\delta^{(1)})^{1+\frac{2}{p}} \left(\frac{(1+2p)}{1 - \left(s - \frac{2}{p}\right)} \right). \quad (5)$$

Notably, the displacement, δ , has the superscript of (1), indicating that it is only the value from the largest fault.

Finally, the amount to radial contraction can be derived by calculating the initial radius of Mercury, R_i , from the fault population strain in Equation 5 via the relation presented by (Watters & Nimmo, 2010):

$$R_i = \left(\frac{R_f^2}{e_{3,t}^{\text{tot}} + 1} \right)^{0.5}, \quad (6)$$

where R_f is the final radius, which we take to be Mercury's current mean radius of 2,440 km. The change in radius, corresponding to the radius decrease from global contraction, is then calculated as $\Delta R = R_f - R_i$.

Table 1
Parameters Used to Solve for Fault Strain

Symbol	Definition	Value
$\mathbb{P}_{3,t}^{(i)}$	The probability for an individual fault to be intersected by a random line parallel to t	—
α	Fault dip angle.	$\alpha = 22^\circ, 30^\circ, \text{ or } 40^\circ$ ^a
θ	Angle between the normal vector to the fault plane and the horizontal vector t .	$\alpha = 22^\circ; \theta = 68^\circ$ $\alpha = 30^\circ; \theta = 60^\circ$ $\alpha = 40^\circ; \theta = 50^\circ$
ϕ	Angle between the horizontal vector t and the slip direction of the fault.	$\alpha = 22^\circ; \phi = 158^\circ$ $\alpha = 30^\circ; \phi = 150^\circ$ $\alpha = 40^\circ; \phi = 140^\circ$
$\delta^{(1)}$	Maximum displacement of the largest fault in the faulted volume. If the superscript (1) is replaced with (i) , this would be the displacement of the i th fault in the faulted volume.	$\delta^{(1)} = 9,300 \text{ m}$ ^a
λ	Geometric shape factor defining shape of the fault plane as either rectangular or elliptical.	Rectangular faults; $\lambda = 0.41$ ^a Elliptical faults; $\lambda = 0.32$
A_f^i	The area of the i th fault plane.	—
A_t	The total cross-sectional area of the faulted volume normal to t .	—
B	Inverse of the scaling factor of the displacement-to-length relationship.	$B = 87.98 \text{ (m)}^p/\text{m}$ ^a $\alpha = 22^\circ; B = 337.9 \text{ (m)}^p/\text{m}$ ^b $\alpha = 30^\circ; B = 451.0 \text{ (m)}^p/\text{m}$ ^b $\alpha = 40^\circ; B = 579.8 \text{ (m)}^p/\text{m}$ ^b $\alpha = 22^\circ; B = 101.9 \text{ (m)}^p/\text{m}$ ^c $\alpha = 30^\circ; B = 136.0 \text{ (m)}^p/\text{m}$ ^c $\alpha = 40^\circ; B = 174.9 \text{ (m)}^p/\text{m}$ ^c
D	Thickness of the faulted volume.	$D = 30 \text{ km}, 40 \text{ km}, \text{ or } 50 \text{ km}$ ^a
$e_{3,t}^x$	Three-dimensional strain as denoted by the subscript 3 in the direction t . When $x = \text{tot}$ it is the total strain of the faulted volume, $x = (1)$ it is the strain of the i th fault, and $x = (1)$ it is the strain associated with the largest fault.	—
p	Exponent of the power law displacement-to-length relationship.	$p = 0.97$ ^a $p = 1.11$ ^b $p = 1.00$ ^c
m	Additive inverse of the exponent of the cumulative length-frequency relationship.	$m = 1.94$ ^a $m = 2.61$ ^b $m = 2.05$ ^c
s	Ratio of m to p .	$s = 2.00$ ^a $s = 2.35$ ^b $s = 2.05$ ^c
T	The dimensional length of the faulted volume in direction t .	—

Note. Statistical parameters B , p , s , and m were derived from regressions performed in this work using measurements published in the original data sets. ^aValue taken or derived from data published in Loveless et al. (2024, 2025a, 2025b). ^bValue derived from data published in Byrne et al. (2014). ^cValue derived from data published in Watters (2021).

3. Parameterization

We use the data from Byrne et al. (2014), Watters (2021), and Loveless et al. (2024, 2025a, 2025b) to parameterize Equation 5 with all required fault-geometric values (Table 1). Geometric forward modeling of 55 shortening landforms on Mercury (Loveless et al., 2025a) yielded thrust fault parameters that provide detailed observational constraints, including ranges of values for fault dips and fault aspect ratios needed here to calculate the geometric shape factor. In particular, Loveless et al. (2025a) establish that the average ranges of fault dip on Mercury are between 22° and 40° , a somewhat wider range than those assumed in the previous global

contractional strain analyses of Mercury (e.g., Byrne et al., 2014; Di Achille et al., 2012; Watters, 2021). The average fault dip of 22° reported in Loveless et al. (2025a) incorporates the entire listric nature of the modeled thrust faults, whereas the maximum dip average of 40° was found to occur along the portion of the fault that had experienced the greatest amount of slip. We therefore calculate the fault strain for dip values bounded by 22° and 40°, as well as the case for rock-mechanically calculated optimal dips of 30° (e.g., Jaeger et al., 2007, p. 91; Schultz, 2019, p. 93). This value is also found to match well with fault dip calculations from structural observations, such as those in Pegg et al. (2021) who reported $28^\circ \pm 5^\circ$. Additionally, the value of 30° has been used as the median fault dip for all previous fault strain calculations (e.g., Byrne et al., 2014; Di Achille et al., 2012; Watters, 2021; Watters & Nimmo, 2010), and thus is not only widely considered the optimal dip value of Mercury's thrust faults but also allows for direct comparisons between our and the previous calculations.

The average depth extent of faulting was determined to be 22 km with the largest structures to be no deeper than ~50 km (Loveless et al., 2025a), and thus we use values of 30 km, 40 km, and 50 km in our calculations to constrain the thickness of the faulted volume, which agrees well with lithospheric thickness and crustal depth estimates (e.g., Padovan et al., 2015). In addition, several other thrust fault modeling efforts have reported maximum penetration depths of up to 35–40 km (e.g., Egea-Gonzalez et al., 2012; Peterson et al., 2020; Watters et al., 2001, 2002). Therefore, we use a value of 40 km as a representative thickness of the faulted volume because the vast majority of fault models return faults with shallower depths and only a handful faults exceed depths of 40 km (Loveless et al., 2025a). This thickness is not likely to be constant throughout Mercury, and a thickness of 40 km likely averages out any variations in the faulted volume.

We parameterize the shape factor λ by taking the average aspect ratio derived in Loveless et al. (2025a) as the ratio of the modeled down dip length of the fault plane to the map-view length of the fault (Table 1). This value was found to be 0.41 and is thus equal to λ if all faults are assumed to be rectangular. If faults are assumed to have an elliptical planar geometry, then we multiply this aspect ratio by $\pi/4$, providing a value of $\lambda = 0.32$. The aspect ratio was derived from listric fault geometries modeled in Loveless et al. (2025a). It is defined only by the horizontal and downward dipping dimensions of the fault plane and not by variations in dip along the fault plane (i.e., listric or homoclinal). We note that the block diagram in Figure 2 shows a rectangular shape of a homoclinal plane for simplicity. We calculate radial contraction for both end-member cases, assuming all faults are either rectangular or elliptical.

Enterprise Rupes is the largest shortening landform on Mercury both in mapped length and vertical relief (Byrne et al., 2018; Ferrari et al., 2015; Watters et al., 2016) and thus is likely underlain by the thrust fault accommodating the largest displacement on Mercury. In order to determine the maximum displacement of the fault, Loveless et al. (2025a) constructed a 2D balanced cross-section of the fault system along the location of maximum relief, matching observed topography, shortening strain from folding, and the slip vector that was indicated by offset impact craters. It was found that Enterprise Rupes consists of two oppositely dipping thrust faults with the primary thrust showing a displacement of value of $\delta^{(1)} = 9,300$ m.

Values B and p in Equation 5 are identified from linear regressions of the power-law displacement-to-length relationship of the population of faults (Cowie & Scholz, 1992a):

$$\delta = \frac{1}{B} L^p. \quad (7)$$

Many works have shown that the displacement-length relationship of thrust faults on Mercury is nearly linear (e.g., Byrne et al., 2014; Di Achille et al., 2012; Schultz et al., 2006; Watters, 2021; Watters & Nimmo, 2010), as is what is observed on Earth (e.g., Schultz, 2019, p. 450). Previous studies use model 1 regressions, a type of linear regression, to derive a scaling relationship, and while assuming a value of 1 for p , produce slope values for $\frac{1}{B}$ (often referred to as γ in other studies) typically on the order of $\sim 6\text{--}9.5 \times 10^{-3}$ (e.g., Byrne et al., 2014; Di Achille et al., 2012; Watters, 2021; Watters & Nimmo, 2010). To provide statistically unbiased results for the regression parameters, we apply a model 2 linear regression to all data sets, as the independent variables (x values) of the regression, that is, the fault length values, are not taken at controlled intervals, which is an assumption of a model 1 regression (Sokal and Rohlf, 1995).

The fault displacements in Byrne et al. (2014) and Watters (2021) were found by relating the observed structural relief via the fault dip for a truncated selection of the sample, where dips were assumed to be 25°, 30°, and 35°. We

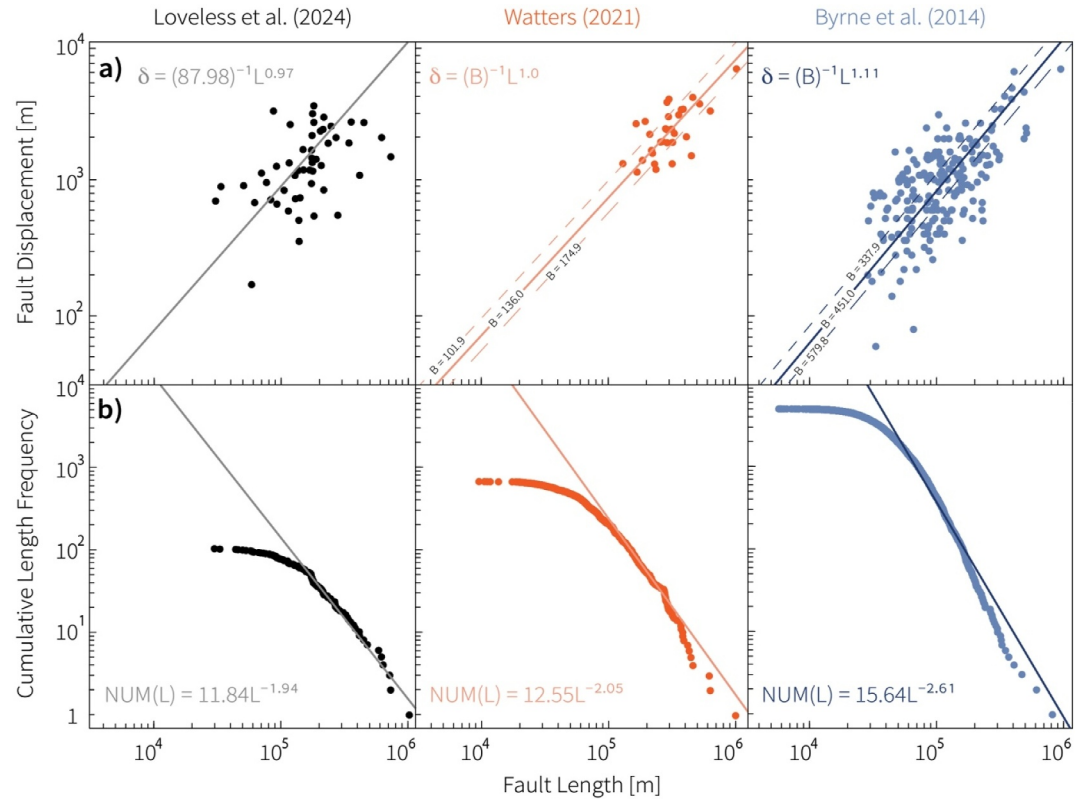


Figure 3. Thrust fault population statistics for three data sets on Mercury, where the data by Loveless et al. (2024, 2025a, 2025b), Watters (2021), and Byrne et al. (2014) are shown in black, orange, and blue in the left, center, and right columns, respectively. (a) Fault displacement-to-length data and the corresponding regressions shown for all data sets in the top row. Data (points) and regression (solid line) for Loveless et al. (2024) were derived from model results (Loveless et al., 2025a), hence showing only one regression line and one value of B in this panel. Data (points) from Byrne et al. (2014) and Watters (2021) are shown for the assumption that all measurements are from faults with dips of $\alpha = 30^\circ$. Regressions for those data are shown for $\alpha = 30^\circ$ with a solid line and for $\alpha = 22^\circ$ and 40° with dashed lines, respectively. The different dip angles produce different displacement data and therefore create different values of B for the Watters (2021) and Byrne et al. (2014) data sets. All three data sets utilized truncated samples of their original shortening landform data sets for displacement data: Loveless et al. (2025a) modeled 55 of the 100 shortening landforms mapped in Loveless et al. (2024), Watters (2021) collected relief data for 31 of their 653 shortening landforms, and Byrne et al. (2014) collected relief data for 216 of their 5,934 shortening landforms. (b) Cumulative fault length frequency distributions shown for all data sets in the bottom row. Cumulative length-frequency regressions are shown as solid lines for the linear portion of the data distribution. Color-coding of the data sets corresponds to that in Figure 1.

carry out three model 2 regressions for these two data sets to account for the updated range of dip angles (Loveless et al., 2025a). The Byrne et al. (2014) data set provided relief data for 216 of the 5,934 shortening landforms. The Watters (2021) data set provided relief data for 31 of the 653 shortening landforms. We show those regression results in Figure 3a along with displacement-to-length data derived from 55 balanced cross-sections of thrust fault systems in Loveless et al. (2025a). From these regressions, we extracted the values p and B and listed their values in Table 1. Because no fault dip values are known in the structural relief measurements by Byrne et al. (2014) and Watters (2021), the three different assumptions of dips produce three different values for parameter B . The different methods that the three data sets use to estimate fault displacements may influence the values for p and B . The values for parameter p remain unchanged for different assumptions of fault dip values.

The value for s used in Equation 5 is derived from the slope, $-m$, of the power-law length-frequency distribution, $\text{NUM}(L)$, of the mapped fault population (Cladouhos & Marrett, 1996; Watterson et al., 1996) as follows:

$$\text{NUM}(L) = G L^{-m}; s = \frac{m}{p}. \quad (8)$$

Table 2
Radius Change Values Listed for Those Parameters That Have the Largest Impact on Calculations

	D [km]	α [°]	ΔR [km] Loveless et al. (2024, 2025a, 2025b) ($n = 100$)	ΔR [km] Watters (2021) ($n = 653$)	ΔR [km] Byrne et al. (2014) ($n = 5,934$)
Minimum	50	22	1.5–1.9	0.9–1.1	1.3–1.5
Optimal	40	30	2.3–2.9	2.4–3.0	2.8–3.5
Maximum	30	40	3.5–4.4	6.0–7.6	6.6–8.4

Note. The lower and upper bounds of each of the ranges listed for ΔR represent elliptical and rectangular shapes of the fault tip line, respectively. Bold numbers are the Optimal ΔR values.

Here, G is the scaling factor relating the cumulative number of faults to the fault length. It is equal to the cumulative number of faults that have a unit length (Twiss & Marrett, 2010a). Because extrapolation over the total spread of the cumulative length frequency distribution produces overestimates of the strain, we follow practices laid out in Twiss and Marrett (2010a) and apply a regression to only its linear portion (Figure 3b). For each data set, we extract the values for m , calculate s (Equation 8) and list the results in Table 1. We note that the displacement-to-length data for all three data sets (Figure 3a) is a subset of the fault sample size (Figure 3b), producing a different number of data points between the two graphs.

4. Results: Radial Contraction Estimates

We present our results of radial contraction for the combination of possible geometric properties in Table 2. Overall, our analysis reveals that the amount of radial contraction on Mercury is anywhere between 1.5 and 4.4 km for the Loveless et al. (2024, 2025a, 2025b) data set, 0.9 and 7.6 km for the Watters (2021) data set, or 1.3 and 8.4 km for the Byrne et al. (2014) data set. These total ranges depend on the combination of the thickness of the faulted volume, fault dips, and shapes of the fault plane. Thinner faulted volumes, higher fault dips, and rectangular fault shapes produce higher amounts of strain when compared to thicker faulted volumes, gentler fault dips, and elliptical fault shapes, respectively.

In particular, the deeper we assume that the faulted volume extends, the larger the volume hosting the fault population, thus lowering the volumetric strain. With decreasing dip, the one-dimensional strain of the largest fault increases (i.e., the term in brackets in Equation 2), but the proportion of the projected area of the fault plane onto the cross-sectional area of the faulted volume goes down, and thus the probability, $\mathcal{P}$, goes down (Equation 3). Therefore, the strain calculated in one dimension follows the trend in Byrne et al. (2014) and Watters (2021) where strain estimates increase with lower fault dips. However, the probability governs the total outcome of Equation 5, and therefore, explains why lower amounts of strain are produced for lower dip angles. Rectangular shapes of the fault tip line create a larger area of fault planes to be projected onto the cross-sectional area of the faulted volume than elliptical shapes. Therefore, the calculated amount of strain is larger for rectangularly shaped fault tip lines.

We find a combination of parameters that produce a minimal amount of global contraction ($\Delta R_{\min}$), a combination of optimal parameters that produces the most probable amount of global contraction (Optimal ΔR), and a combination of parameters that produces a maximum amount of global contraction ($\Delta R_{\max}$) from the investigated fault population data sets. We find that all three ranges for ΔR are similar across the data sets. The combination of parameters that produces $\Delta R_{\min}$ includes an average thickness of the faulted volume of 50 km, with all faults in the population dipping 22° (Table 2). This set of parameters produced a $\Delta R_{\min}$ of 1.5–1.9, 0.9–1.1, and 1.3–1.5 km for the Loveless et al. (2024, 2025a, 2025b), Watters (2021), and Byrne et al. (2014) data sets respectively. The lower and upper bounds for these ranges of ΔR represent elliptical and rectangular shapes of the fault tip line, respectively.

While dip values of 30° and a thickness of the faulted volume of 40 km imply that they are assumed to be constant for the fault population and throughout Mercury, they also represent conservative estimates of averages, as there are established ranges of variations of these parameters around these values. Therefore, we consider the resulting ΔR values (bold, Table 2) as being optimal, producing ΔR of 2.3–2.9, 2.4–3.0, and 2.8–3.5 km for the Loveless et al. (2024, 2025a, 2025b), Watters (2021), and Byrne et al. (2014) data sets respectively.

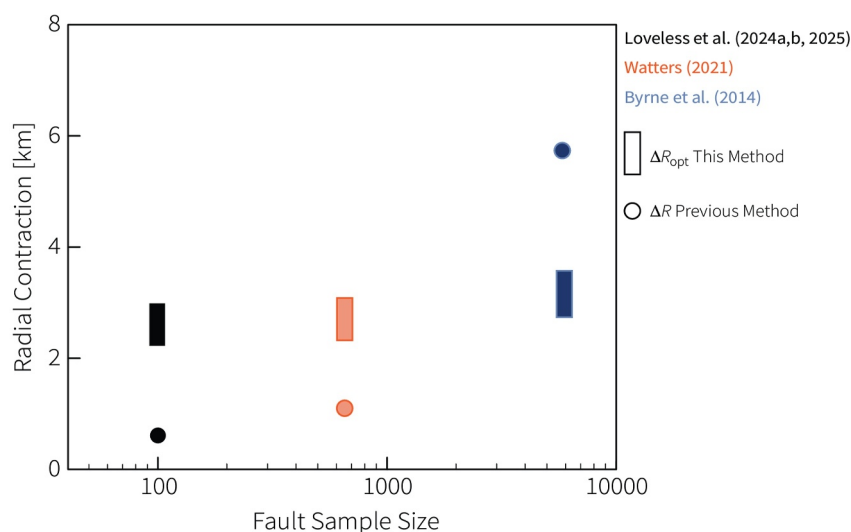


Figure 4. Radial contraction estimates for Mercury as a function of fault sample size. Circles represent radial contraction estimates assuming all faults dip at 30° using the method described in Watters (2021). Rectangles represent the range of results considered for optimal parameters in this study. Colors correspond to those in Figures 1 and 3.

A maximum amount of radial contraction is calculated by assuming a thickness of the faulted volume of 30 km depth and that all faults in the fault population dip 40° . With this combination of fault population parameters, the Loveless et al. (2024, 2025a, 2025b) data set returns a $\Delta R_{\max}$ of 3.5–4.4 km. The Watters (2021) and Byrne et al. (2014) data sets produced $\Delta R_{\max}$ of 6.0–7.6 and 6.6–8.4 km, respectively (Table 2).

5. Discussion and Conclusions

Our results establish that each data set, irrespective of the number of considered faults, produces similar amounts of contractional strain, which, in turn, yields radius change estimates that are comparable to one another, as shown for parameters considered optimal in Figure 4 (rectangles). We show this range of values in comparison to the optimal cases of fault dips of 30° using the previous method (circles, Figure 4). Note that there is no radius change value reported for Loveless et al. (2024) using the previous method, but we calculated it here for completeness. Figure 4 highlights the dependency on the number of faults considered in the previous method and the advantage of the method presented in this study. This underscores the utility of this methodology for estimating tectonic strain on any planetary bodies where fault populations can be observed.

In particular, for the Watters (2021) data set, we find radial contraction estimates of 0.9–7.6 km, with the optimal range of ΔR being 2.4–3.0 km. Watters (2021) calculated substantially smaller values of ΔR , amounting to 0.9, 1.1, and 1.3 km for $\alpha = 35^\circ$, 30° , and 25° , respectively. These values appear to agree with our $\Delta R_{\min}$ values across all three data sets (Table 2). However, the $\Delta R_{\min}$ values in this work represent the extreme lower bounds of physical parameters, whereas the range of radius change values produced in Watters (2021) spans the entire considered parameter space in that publication. Therefore, the $\Delta R_{\min}$ results presented here cannot be considered to confirm the range of estimates presented in Watters (2021). The low values for Mercury's contraction presented in Watters (2021) arise from the low number of shortening landforms considered to have contributed to global contraction.

Using the same method as Watters (2021) but for a larger number of faults, Byrne et al. (2014) estimated Mercury's radial contraction to be ~ 3.1 – 7.1 km, which falls in the range of optimal ΔR to $\Delta R_{\max}$ calculated in this study (Table 2). The larger sample size is likely closer to the true number of faults in Mercury's population of thrust faults, which likely explains their results showing a better overlap with ours. The large sample size of this data set and the sampling of structural reliefs over a wide range of fault lengths—as seen in the spread of the displacement-to-length data when compared to the length frequency distribution (Figure 3, right column)—makes this data set statistically more probable. In contrast, the data set by Watters (2021) sampled the structural relief of only the largest of shortening landforms, which is seen in the spread of the data being skewed toward the longest faults (Figure 3, center column).

Although ΔR values derived in this study for all three data sets across the entire parameter space agree well with each other, some differences between the radial contraction estimates derived from the Loveless et al. (2024, 2025a, 2025b) data set with the Byrne et al. (2014) and Watters (2021) data sets arise (Table 2). These differences, however, are much smaller than those using the previous method and they result from different values of the parameter B , which is due to the different structural relief measurements and the displacement statistics that are derived from them (Figure 2a).

As mentioned above, the ΔR estimates in both methods assume that all considered faults have the same geometric properties. As faults have a self-similar growth (Cowie & Scholz, 1992b), this is a reasonable assumption for the geometric shape factor. However, it is improbable that all faults within a population have the same dip, as such Mercury's faults are found to widely vary in dip (Loveless et al., 2025a). It is equally improbable that all dips would be solely composed of values at either the upper or lower bound of our range. With respect to the entire fault population, strain calculations for the optimal dip values would likely produce results that are most representative of the natural variation of dips. Additionally, the thickness of the faulted volume is similarly likely to vary across Mercury. Therefore, a value between the minimum and maximum observed values at 40 km would yield results accounting for the natural variation of this parameter, too. Thus, we consider the contraction estimates for optimal ΔR to likely best capture the amount of global contraction on Mercury.

We find Mercury's radius change that can be attributed to thrust faulting to show optimal values between 2.3 and 3.5 km consistently across all three data sets (Figure 4). Accounting for the additional component of global contraction that could have taken place (Klimczak, 2015), 0.4–2.1 km of radial contraction may be considered to be added to any radial contraction estimates from faulting. Adding these two components yields a range of radial contraction of 2.7–5.6 km that is likely to have occurred on Mercury. This result is in reasonable agreement with many thermal evolution modeling efforts (e.g., Breuer et al., 2007; Hauck et al., 2004; Michel et al., 2013; Solomon, 1977) and, in combination with timing and strain rate estimates of global contraction (Crane & Klimczak, 2017), may be used to tightly constrain future thermal evolution modeling.

In this work, we adapted the methodology presented and tested by Twiss and Marrett (2010a, 2010b) to estimate the amount of thrust fault strain Mercury experienced from global contraction. Using three data sets sampling vastly different numbers of faults of the same population (Figure 1), we demonstrated that any bias introduced by incompletely sampled fault populations using the previous method is avoided using the method presented here. Thus, any future calculations of Mercury's radial contraction, including those that separate different fault populations based on region or geology, will produce more robust estimates using the framework of Twiss and Marrett (2010a). We also suggest that this methodology can be used to calculate fault population strain for other planetary processes and bodies.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

We did not use any new data in this research. All of our data came from previously published sources. Data is available through Byrne et al. (2014); Watters (2021); and Loveless et al. (2024, 2025a, 2025b). Our code in R used for our calculations is available on Mendeley Data at: <https://data.mendeley.com/datasets/htkkrpc9w5/1> (Loveless & Klimczak, 2025).

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